

# Targeted therapy for chronic lymphocytic leukaemia (CLL)

Targeted therapies are drugs that target something that is helping the cancer cells grow and survive. Most people with CLL are treated with a combination of targeted therapy and chemotherapy.

## On this page

- [What are targeted therapies?](#)
- [Targeted therapies for chronic lymphocytic leukaemia \(CLL\)](#)
- [B-cell receptor \(BCR\) inhibitors](#)
- [BCL2 inhibitors](#)
- [Monoclonal antibodies](#)
- [Side effects of targeted therapies for chronic lymphocytic leukaemia \(CLL\)](#)
- [About our information](#)
- [How we can help](#)

## What are targeted therapies?

Targeted therapy uses drugs to find and attack cancer cells. There are many different types of targeted therapy. Each type targets something in or around the cancer cell that is helping it grow and survive. Because these drugs only target the cancer cells, they have less effect on healthy cells than other treatments.

# Targeted therapies for chronic lymphocytic leukaemia (CLL)

Most people with [chronic lymphocytic leukaemia \(CLL\)](#) are treated with a combination of targeted therapy and [chemotherapy](#).

There are different types of targeted therapy. They all act in different ways and have different side effects. To decide which treatments may be suitable for you, your [haematologist](#) will look at:

- the [stage](#) of the leukaemia
- your test results
- your general health
- your [gene tests](#), and whether you have a p53 mutation.

## B-cell receptor (BCR) inhibitors

BCR inhibitors are a type of targeted therapy used to treat CLL. They work by targeting proteins on the surface of cancer cells. The B-cell receptor is a protein found on the surface of CLL cells. It controls how CLL cells grow. If the B-cell receptor is switched off, the CLL cells die.

BCR inhibitors target the B-cell receptors to stop the cancer growing and dividing.

The drugs that can be used for CLL are:

- [ibrutinib](#) (Imbruvica<sup>®</sup>)
- [acalabrutinib](#).

You take these as tablets or capsules every day. You usually continue taking them for as long as they are working.

## BCL2 inhibitors

One of the reasons that CLL cells continue to grow is that they make too much of a protein called BCL2. BCL2 inhibitors work by blocking (inhibiting) this protein. This helps destroy the CLL cells.

Your haematologist may suggest treatment with a BCL2 inhibitor called [venetoclax](#) (Venclyxto<sup>®</sup>). If you have not had previous treatment for CLL, it is given with the monoclonal antibody obinutuzumab. If you have had previous treatment for CLL, you may have venetoclax with a monoclonal antibody called rituximab.

You take venetoclax as tablets. You may have it with other treatments. This drug may only be available in some situations. Your haematologist can tell you if it is suitable for you.

# Monoclonal antibodies

Monoclonal antibodies treat CLL by targeting proteins on white blood cells called B-lymphocytes (B-cells). This makes the B-cells die.

Monoclonal antibodies are usually given with chemotherapy or with other types of targeted therapy. They are usually given as a drip into a vein.

Monoclonal antibody treatments used for CLL include:

- [rituximab](#)
- [obinutuzumab](#) (Gazyvaro<sup>®</sup>)
- ofatumumab (Arzerra<sup>®</sup>).
- [alemtuzumab](#) – this is only sometimes used to treat CLL.

## How monoclonal antibodies are given

Monoclonal antibodies are usually given into a vein as a drip (infusion). You can have them as an outpatient. You may be asked to stay in hospital overnight for your first treatment.

You may have a reaction during the infusion. A reaction is more likely with the first infusion, so it is given slowly over a few hours. Reactions are usually milder after the first infusion. You are also given medicines before each infusion to help prevent or reduce any reaction.

Symptoms of a reaction can include:

- a high temperature
- shivering and shaking
- a rash
- low blood pressure
- [feeling sick](#) (nausea).

If you have any of these symptoms or feel unwell during or soon after an infusion, tell your nurse or doctor straight away.

Alemtuzumab can also be given as an injection under the skin (subcutaneously). In this case, it is less likely to cause a reaction. But it may cause some redness, swelling or soreness around the area where it is injected. This usually goes away after 1 to 2 weeks.

## Side effects of targeted therapies for chronic lymphocytic leukaemia (CLL)

Some side effects of targeted therapies may include:

- risk of infection
- [diarrhoea](#)

- bruising and bleeding
- [feeling sick](#)
- rarely, tumour lysis syndrome (TLS).

Different targeted therapies can have different side effects. Your doctor, nurse or pharmacist will explain the possible side effects of the [specific drugs](#) you are having before you start treatment.

Your nurse will give you phone numbers for the hospital. If you feel unwell or need advice, you can call them at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

## About our information

### References



Below is a sample of the sources used in our chronic lymphocytic leukaemia (CLL) information. If you would like more information about the sources we use, please contact us at [cancerinformationteam@macmillan.org.uk](mailto:cancerinformationteam@macmillan.org.uk)

Schuh A et al. Guideline for the treatment of chronic lymphocytic leukaemia. British Society for Haematology Guidelines. July 2018.

Eichhorst B et al. Chronic lymphocytic leukaemia: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. ESMO Guidelines Committee. October 2020.

### Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Dr Anne Parker](#), Consultant Haematologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

## The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

## Date reviewed

Reviewed: 01 February 2022 | Next review: 01 February 2025

ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

## How we can help

### Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



#### Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



#### Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

#### Chat online



Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to

## **Online Community**

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



## **Find cancer support services in your area**

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

**Get help**